

BrewUp Production Operations Manual

Introduction

This document describes the brewing and production processes used by BrewUp Brewery.

The purpose of this manual is to provide production personnel, warehouse operators, and management teams with a consistent reference for beer manufacturing activities.

The Production department is responsible for transforming raw ingredients into finished beer products while maintaining quality, consistency, and operational efficiency.

Production Responsibilities

The Production department is responsible for:

- Brewing operations
- Recipe execution
- Fermentation management
- Quality control
- Packaging coordination
- Production scheduling
- Batch traceability

Production personnel must follow documented procedures and record all relevant production information in the BrewUp ERP system.

Brewing Process Overview

The brewing process consists of the following stages:

1. Raw Material Preparation
2. Mashing
3. Lautering
4. Boiling
5. Whirlpool
6. Cooling
7. Fermentation
8. Conditioning
9. Packaging
10. Quality Validation

Each stage must be completed according to the approved recipe specifications.

Raw Materials

Malt

Malt provides fermentable sugars and contributes flavor, body, and color.

Common malts used by BrewUp include:

- Pale Ale Malt
- Pilsner Malt
- Munich Malt
- Wheat Malt
- Crystal Malt

All malt deliveries must be inspected before acceptance.

Hops

Hops contribute bitterness, flavor, and aroma.

Frequently used hop varieties include:

- Cascade
- Centennial
- Citra
- Mosaic
- Simcoe
- Amarillo

Hop inventory must be stored in temperature-controlled environments.

Yeast

Yeast converts fermentable sugars into alcohol and carbon dioxide.

Production personnel must verify:

- Yeast viability
- Storage temperature
- Expiration date

Yeast health is considered a critical success factor.

Mashing Operations

The purpose of mashing is to convert starches into fermentable sugars.

Typical mash parameters:

- Temperature: 65°C to 67°C
- Duration: 60 minutes
- Water-to-grain ratio: according to recipe

Operators must record:

- Mash start time
- Mash end time
- Actual temperature profile

Temperature deviations greater than 2°C require supervisor review.

Boiling Operations

Boiling sterilizes the wort and extracts hop bitterness.

Standard boil duration:

- 60 minutes

Hop additions are categorized as:

Bittering Addition

Added at the beginning of the boil.

Purpose:

- Bitterness extraction

Flavor Addition

Added during the middle phase.

Purpose:

- Flavor contribution

Aroma Addition

Added near the end of the boil.

Purpose:

- Aroma preservation

All additions must be recorded in the production batch log.

Fermentation Management

Fermentation begins immediately after cooling.

Production personnel must monitor:

- Temperature
- Specific gravity
- Fermentation progress
- Yeast performance

Recommended fermentation temperatures:

IPA

18°C - 21°C

Weiss

17°C - 20°C

Stout

19°C - 22°C

Unexpected fermentation behavior must be escalated to the Production Manager.

Dry Hopping Procedure

Dry hopping is primarily used for hop-forward styles such as IPA.

Procedure:

1. Complete primary fermentation.
2. Verify target gravity.
3. Add hops according to recipe.
4. Minimize oxygen exposure.
5. Maintain contact time between 3 and 7 days.

Common risks:

- Oxidation
 - Hop creep
 - Contamination
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Quality Control

Quality control activities occur throughout production.

Measurements include:

- Alcohol by Volume (ABV)
- Bitterness (IBU)

- Color (SRM)
- pH
- Specific Gravity

Quality failures must be documented and investigated.

Packaging Operations

Packaging may involve:

- Bottling
- Kegging
- Canning

Before packaging:

- Verify product quality
- Verify carbonation levels
- Verify packaging materials

Packaging records must be retained for traceability purposes.

Production Planning

Production planning is coordinated with Warehouse and Sales departments.

Inputs include:

- Current inventory
- Open customer orders
- Reorder thresholds
- Seasonal demand

Production plans are reviewed weekly.

Inventory Replenishment Policy

Production receives replenishment requests from Warehouse.

The standard replenishment workflow is:

1. Warehouse detects inventory below threshold.
2. Warehouse generates a replenishment recommendation.
3. Production evaluates capacity.
4. Production schedules a brewing batch.

5. Warehouse receives replenishment confirmation.

Critical inventory situations receive priority treatment.

Batch Traceability

Every production batch must be traceable.

Required information:

- Batch identifier
- Recipe version
- Production date
- Ingredients used
- Operators involved
- Quality measurements

Traceability data must be retained for audit purposes.

Safety Guidelines

Production personnel must:

- Wear protective equipment.
- Follow cleaning procedures.
- Follow chemical handling guidelines.
- Report incidents immediately.

Safety violations must be escalated to management.

Continuous Improvement

BrewUp promotes continuous improvement through:

- Process reviews
- Production retrospectives
- Quality trend analysis
- Operational metrics

Improvement initiatives should focus on:

- Consistency
- Quality
- Efficiency

- Waste reduction

The ultimate goal is to produce high-quality beer while maintaining operational excellence.